

Laplace Transform Pairs and Properties

	$f(t)$	$F(s)$
1	$\delta(t)$ - Impulse	1
2	$\mu(t)$ - Step	$\frac{1}{s}$
3	t^n ($n = 1, 2, 3, \dots$)	$\frac{n!}{s^{n+1}}$
4	e^{-at}	$\frac{1}{s+a}$
5	$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
6	$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$

Table 1: Laplace Transform Pairs

	$f(t)$	$F(s)$
1	$Af(t)$	$AF(s)$
2	$f_1(t) \pm f_2(t)$	$F_1(s) \pm F_2(s)$
3	$\frac{d^n}{dt^n} f(t)$	$s^n F(s) - \sum_{k=1}^n s^{n-k} [\frac{d^{k-1}}{dt^{k-1}} f(t)]_{t=0}$
3.1	$\frac{d}{dt} f(t)$	$sF(s) - f(0)$
3.2	$\frac{d^2}{dt^2} f(t)$	$s^2 F(s) - (sf(0) + \dot{f}(0))$
4	$\int_0^t f(t) dt$	$\frac{F(s)}{s}$
5	$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n} F(s), n = 1, 2, 3, \dots$
6	$e^{-at} f(t)$	$F(s+a)$
7	$f(t-a)\mu(t-a)$	$e^{-as} F(s), a \geq 0$
8	$\lim_{t \rightarrow \infty} [f(t)]$	$\lim_{s \rightarrow 0} [sF(s)]$

Table 2: Laplace Transform Properties